

Assignment 2 – Legacy Loan Calculator C++

Student: Syed Mushaf

Course: Software Engineering

Instructor: Engr. Muhammad Umer Haroon

1. Project Overview

The **Legacy Loan Calculator C++** project is a command-line application that calculates EMI (Equated Monthly Installments), total interest, and remaining balances for a loan.

The legacy code had the following issues:

- No input validation
- Hardcoded loan values
- Risk of integer overflow for large tenures
- Monolithic `main.cpp`

Objectives:

1. Fix bugs and validate input
2. Refactor code into modular files
3. Implement unit tests (GoogleTest)
4. Generate Doxygen documentation

Technologies used: C++11, GoogleTest, Doxygen, Git

2. Bug Fixes & Refactoring

2.1 Bug Fixes

Input Validation:

```
if (principal <= 0 || interestRate <= 0 || tenure <= 0) {  
    std::cerr << "Error: Invalid input values.\n";  
    return;  
}
```

Integer Overflow Prevention:

```
long double monthlyRate = interestRate / 1200.0;
long double emi = principal * monthlyRate * pow(1 + monthlyRate, tenure) /
    (pow(1 + monthlyRate, tenure) - 1);
```

Remove Hardcoded Values (from config.txt):

```
principal=100000
interestRate=12
tenure=360
```

```
LoanConfig config = readConfig("config.txt");
```

2.2 Refactoring

- `main.cpp` → CLI & program entry
- `Loan.h` / `Loan.cpp` → Loan class
- `Utils.h` / `Utils.cpp` → helper functions

Benefits: Modularity, maintainability, readability

3. Unit Testing

Test Cases:

Test Case	Description	Result
Normal EMI	Standard EMI calculation	✓ Pass
Invalid Input	Negative/zero values	✓ Pass
Large Tenure	Tenure > 1000 months	✓ Pass

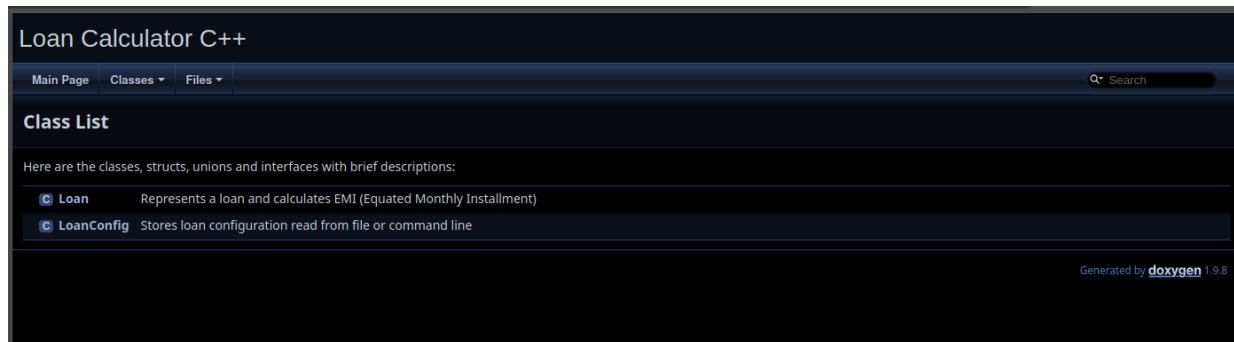
Insert Screenshot of test output here

Example output:

```
syed-mushaf@syed-mushaf-Latitude-5410:~/legacy-calc-2009$ ./test_emi
[=====] Running 3 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 3 tests from LoanTest
[ RUN    ] LoanTest.NormalEMI
[ OK     ] LoanTest.NormalEMI (0 ms)
[ RUN    ] LoanTest.InvalidValues
[ OK     ] LoanTest.InvalidValues (0 ms)
```

4. Doxygen Documentation

- Generated HTML docs in docs/html/
- All classes, structs, and functions documented



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Example Doxygen comment:

```
#ifndef LOAN_H
#define LOAN_H

#include <iostream>

/**
 * @class Loan
 * @brief Represents a loan and calculates EMI (Equated Monthly Installment)
 */
class Loan {
private:
```

5. Git Commit History:

```
syed-mushaf@syed-mushaf-Latitude-5410:~/legacy-calc-2009$ git log --oneline
b315831 (HEAD -> dev, origin/dev) Update README with Doxygen docs and build instructions
12c08b2 Add Doxygen HTML documentation
5621e68 Initial setup: added .gitignore and updated README
80d0228 Added src, test folders, Loan class files, config.txt, and .gitignore
b9dfa16 Initial commit: added project structure and files
347e865 (origin/master, origin/HEAD, master) Added compilation information
dff0427 Nicer formatting on Readme and added better app help text
71f0456 Added readme and additional app help text
c3e0801 Added a Qt Gui and scons compilation. The makefile was generated by Qt.
be10569 Changes to use libCmdLineParser
12ee333 Removed compileCommands.txt
4d09e8b Made methods inline and removed some compiler errors
d99e446 First version
```

Mushaf-syed Add Doxygen HTML documentation 12c08b2 · 1 hour ago 12 Commits		
📁 .vscode	Initial commit: added project structure and files	yesterday
📁 docs/html	Add Doxygen HTML documentation	1 hour ago
📁 src	Initial setup: added .gitignore and updated README	2 hours ago
📁 test	Added src, test folders, Loan class files, config.txt, and .g...	yesterday
📄 .gitignore	Added src, test folders, Loan class files, config.txt, and .g...	yesterday
📄 Doxyfile	Initial setup: added .gitignore and updated README	2 hours ago
📄 LoanCalcQtMainWindow.cpp	Added a Qt Gui and scons compilation. The makefile was...	13 years ago
📄 LoanCalcQtMainWindow.h	Added a Qt Gui and scons compilation. The makefile was...	13 years ago
📄 LoanCalculator.cpp	Added a Qt Gui and scons compilation. The makefile was...	13 years ago
📄 LoanCalculator.h	Added a Qt Gui and scons compilation. The makefile was...	13 years ago
📄 LoanCalculatorMain.cpp	Nicer formatting on Readme and added better app help ...	13 years ago
📄 README	Added compilation information	13 years ago
📄 README.md	Initial setup: added .gitignore and updated README	2 hours ago
📄 SConstruct	Added a Qt Gui and scons compilation. The makefile was...	13 years ago
📄 calc	Initial setup: added .gitignore and updated README	2 hours ago
📄 config.txt	Initial setup: added .gitignore and updated README	2 hours ago
📄 loanCalculatorCpp.pro	Added a Qt Gui and scons compilation. The makefile was...	13 years ago
📄 makefile	Added a Qt Gui and scons compilation. The makefile was...	13 years ago

6. Build & Run Instructions

Build:

```
g++ -std=c++11 src/*.cpp -o calc
```

Run:

```
./calc -cp -a 100000 -i 12 -n 360
```

Doxygen Documentation:

```
-> open docs/html/index.html
```

Run Unit Tests:

```
cd test
```

```
g++ -std=c++11 test_emi.cpp -lgtest -lpthread -o test_emi
```

```
./test_emi
```

7. Conclusion

- Legacy bugs fixed
- Modular and clean code structure
- Unit tests passed
- Full Doxygen documentation available
- Repository ready for submission